

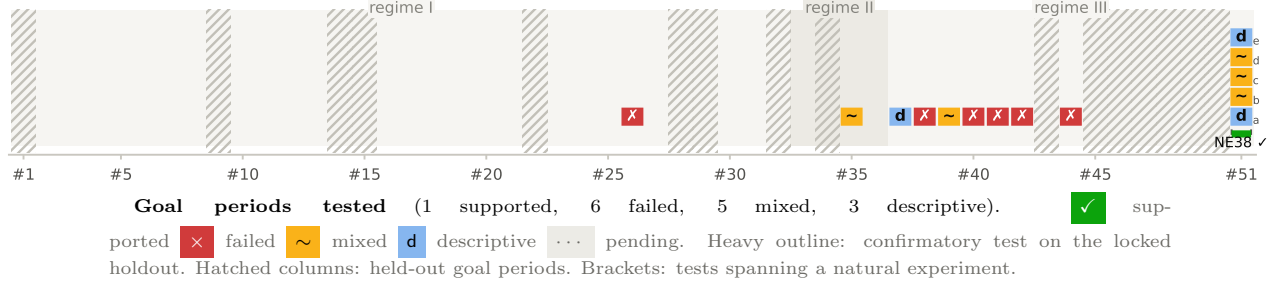
H29: Driver nodes: where an operator message moves the whole swarm

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 50% Faithfulness 1.5/5 Usefulness 2.5/5

Bottom line. Content influence is real and address-gated (named $\gg$ unnamed on the context ledger); the content-pull driver ranking fails, but a reply-graph ranking weakly predicts spread.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Per goal period, a content-influence network measures how far agent j 's next statement moves toward a seen message from i , net of placebos. We rank agents by linear controllability in 9 counted and 3 short non-holdout units (regime III, #51). **Round 1b** repeats everything on context-ledger visibility, with a second embedding model and the DQ2 reply graph. The pre-registered test still fails in the opposite direction ($\kappa < 0$ in 7/9). Messages posted during the recipient's own call are co-responses, not leaked influence. At matched age, influence is real and address-gated. Named recipients move 0.04–0.08 of the gap per message in every #51 segment, unnamed ones 0.00–0.013. The content-pull driver ranking is still noise. A reply-graph ranking (net reply current) is reproducible (split-half 0.65) and weakly predicts held-out spread (pooled $\rho = 0.23$ bge, 0.12 gte). In NE38, an operator's re-assignment of one agent moved it and did not relay. The holdout is not run.

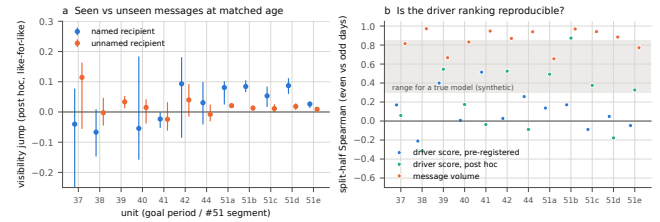
PHYSICAL MODEL

Linear response (02) with vector content states (11): DeGroot / Friedkin–Johnsen dynamics. x_j : the whitened state-embedding of agent j . $V(\tau)$: messages that reached j between its previous and current model calls. a_{ij} : pull per message. $A_{ij} = a_{ij}r_{ij}$, with r_{ij} the messages from i absorbed by j per hour. Γ : relaxation toward the agent's own field. Driver score D_k : reachability of the swarm mean from agent k over $T = 4$ h. $E_k = N^2/D_k$: energy to shift the mean through k .

$$x_j(\tau_n) = x_j(\tau_{n-1}) + \sum_{m \in V(\tau_n)} a_{s(m)j} [x_m - x_j(\tau_{n-1})] + \gamma_j [h_j - x_j] + \xi$$

$$\dot{\mathbf{x}} = \mathbf{Q}\mathbf{x} + \mathbf{e}_k u, \quad Q_{ji} = A_{ij}, \quad Q_{jj} = -\sum_i A_{ij} - \Gamma$$

$$D_k = \int_0^T (\mathbf{1}^\top e^{Q^t} \mathbf{e}_k)^2 dt, \quad \mathbf{1}^\top \mathbf{Q} \mathbf{e}_k = \text{out}_k - \text{in}_k - \Gamma$$



Observables. (a) Post hoc visibility jump at matched age: seen minus unseen messages within 30 s of the reply, like-for-like. Named recipients (blue) vs unnamed (orange). Named $\gg$ unnamed in every #51 segment; the two-room weeks are underpowered. (b) Split-half reproducibility of rankings. Message volume is always reproducible; the driver score only sometimes (grey band: what a true model gives on synthetic data).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

To move one agent, name it: named messages pull 5–25 $\times$ more than unnamed ones in #51, and broadcast content barely propagates. An operator message to one agent stays with that agent (NE38). If you must pick a relay, rank agents by net reply current (replies received minus given), not by volume or content-pull “driver” scores. Expect a weak edge (held-out $\rho \approx 0.1$ –0.2). A designated leader gets $\times 1.39$ replies per message, not more reach (#35). Network controllability adds nothing measurable; the steering-cost scaling ($\propto N^{2.7}$ for a typical agent) is model-only.

